



ST BENEDICT'S

SUBJECT	AP Mathematics	PAPER	Exam - Algebra
GRADE	11	DATE	11 November 2017
EXAMINER NAME	Mr Benecke	MARKS	160
TEACHER		MODERATOR	Mrs Povall & Mrs Sillman
		DURATION	2 hours

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Solving Equations with Real Roots	21	
2	Inequalities	22	
3	Solving Equations with Complex Roots	12	
4	Trigonometry	10	
5	Radian Measure	8	
6	Limits	26	
7	Differentiation	51	
8	Integration	10	
TOTAL		160	

INSTRUCTIONS:

1. This paper consists of 8 questions and 16 pages.
2. Read the questions carefully.
3. Answer all questions.
4. Number your answers clearly and use the same numbering as in the question paper.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. Round off your answers to two decimal digits where necessary.
7. All necessary working details must be shown. Answers only, without the relevant calculations will not be given marks. Equations may not be solved solely with a calculator.
8. It is in your interest to write legibly and present your work neatly.
9. You may detach the Information Sheet at the back of the question paper.

QUESTION 1**21 MARKS****Solve for x . Where $x \in \mathbb{R}$**

a) $\frac{3x+4}{6+x} = \frac{3x-2}{x-3} + \frac{21x}{18-3x-x^2}$ (6)

b) $x^2 - 4\sqrt{x^2 - 2x + 3} = 2x - 3$ (10)

c) $|x + 3| = 2x + 12$

(5)

QUESTION 2

22 MARKS

Solve for x . Where $x \in \mathbb{R}$

a) $\frac{x^2 - 5x - 6}{x^3 - 27} \geq 0$

(7)

b) $\frac{x^2\sqrt{x+2}}{(x+3)(x-4)^2} > 0$ (6)

c) $\frac{2}{x} < \left| \frac{x}{2} \right|$ (9)

QUESTION 3**12 MARKS****Solve for x . Where $x \in \mathbb{C}$**

a) $x^3 + 4x^2 + 25x + 100 = 0$ (6)

b) $x^3 + 2x^2 - x - 14 = 0$, Given that $x = -2 + \sqrt{3}i$ is one of the solutions (6)

QUESTION 4

10 MARKS

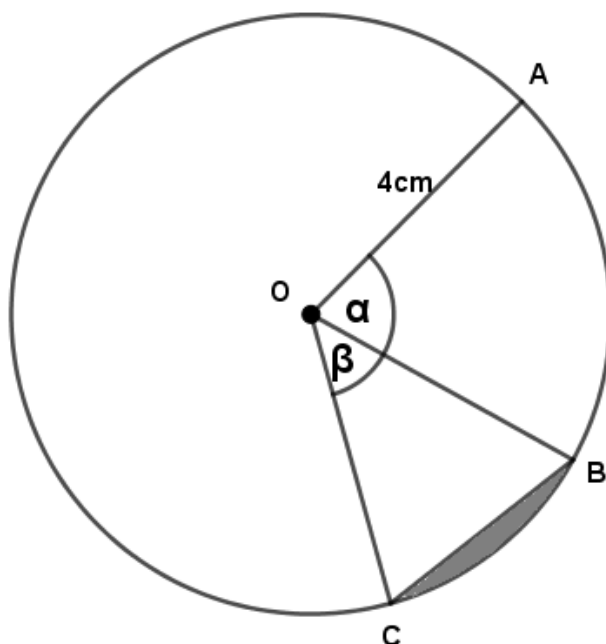
- a) Simplify the following:

$$\frac{1 - \cos^2(x + 90^\circ)}{\operatorname{cosec}^2(180^\circ + x) - 1} \quad (6)$$

- b) If $\sec \theta = -\frac{13}{12}$ and $\sin \theta < 0$, then without the use of a calculator determine the value of $\cot \theta$. (4)

QUESTION 5**8 MARKS**

In the diagram below, the circle with centre O has a radius of 4 cm . The area of the minor sector AOB is $2\pi\text{ cm}^2$ and the length of arc AB is twice the length of arc BC .



- a) Determine the Area of the shaded region.

(8)

QUESTION 6**26 MARKS**

a) Evaluate: $\lim_{x \rightarrow 4} \frac{3x^3 - 12x^2}{x - 4}$ (3)

b) Evaluate: $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$ (5)

c) Evaluate: $\lim_{x \rightarrow \infty} \frac{x^2 + 4x + 4}{4x^2 + 15x + 15}$ (5)

d) Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 7}{2x^3 + 12x}$ (6)

e) Prove that $\lim_{x \rightarrow 4} \frac{|x^2 - 16|}{x - 4}$ does not exist (7)

QUESTION 7

51 MARKS

- a) Determine $f'(x)$ from first principles if $f(x) = 1 - 4x^2$ (7)

- b) Determine the derivative of: (5)

1) $y = \frac{6x^5 + x^3}{2x^2}$

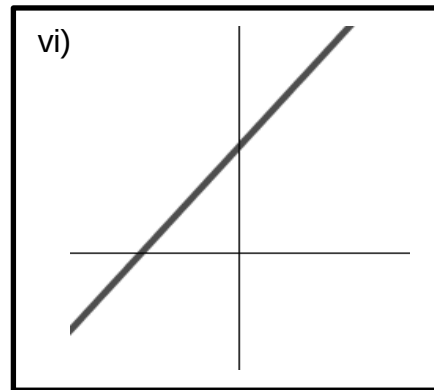
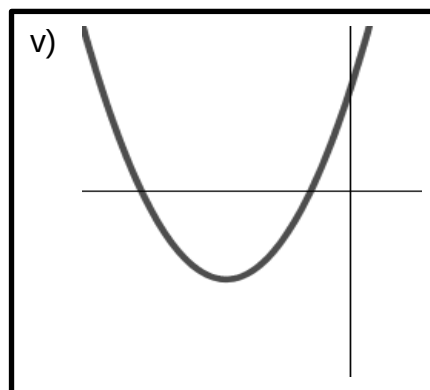
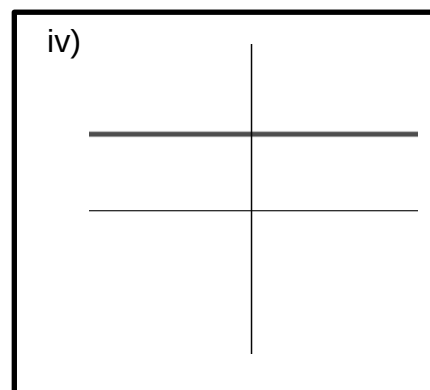
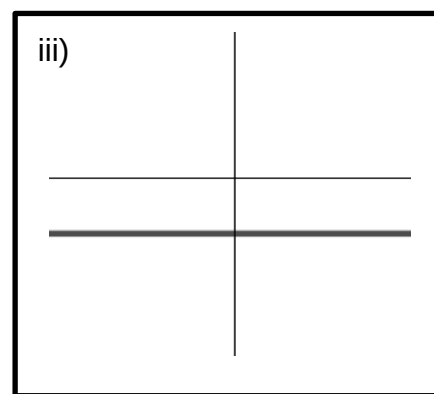
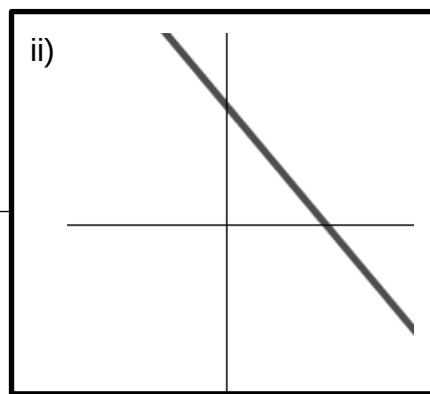
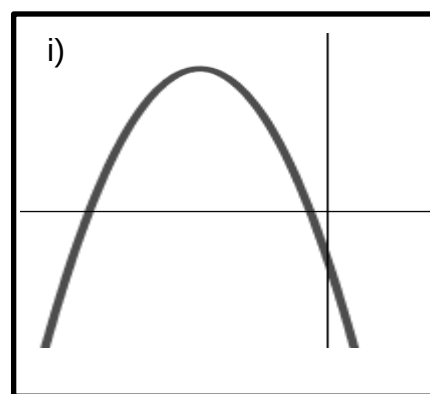
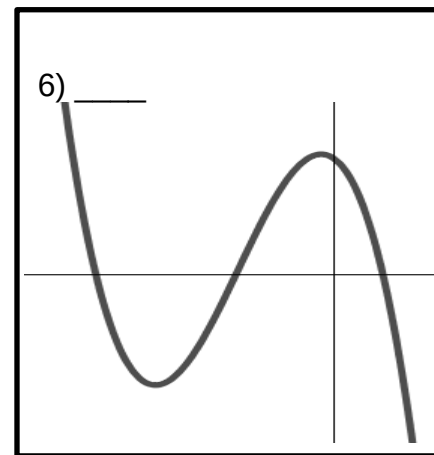
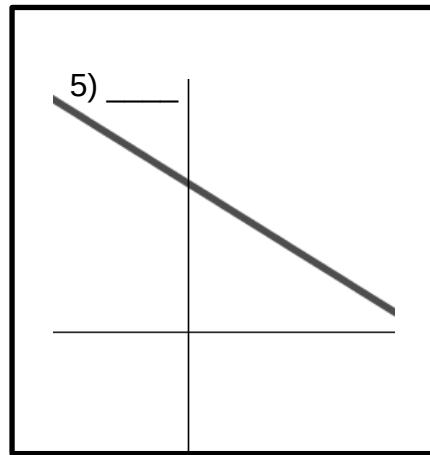
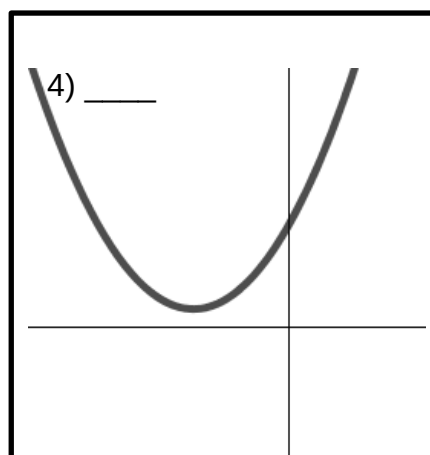
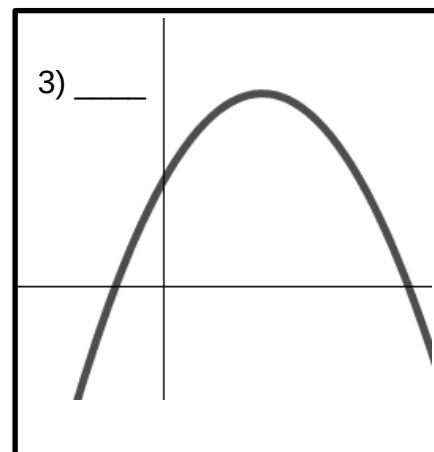
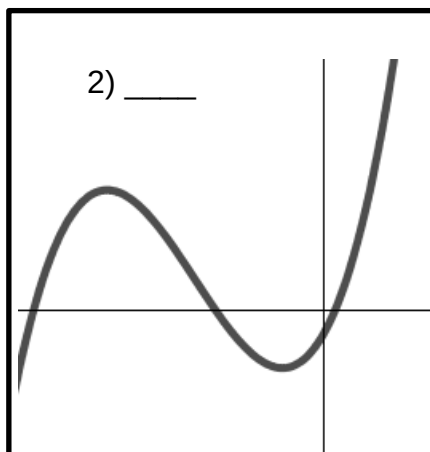
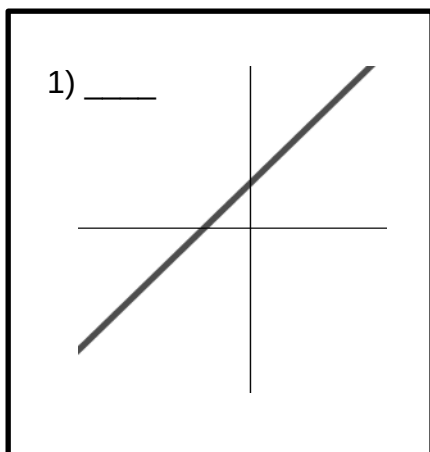
2) $g(x) = \sqrt[3]{x} + \frac{1}{\sqrt{x}}$ (5)

c) Given $f(x) = \frac{1}{3}x^3 + 3x^2 + 5x + 1$

1) Determine the coordinates of the turning points (6)

2) Draw a rough sketch of $f(x)$. (4)

- d) Match each function with its derivative. 1-6 are the functions and i-vi are the derivatives, so write i, ii, iii, iv, v, and vi on the blank line next to the function it matches. (12)



e) Consider the following function:

$$\text{Let } f(x) = \begin{cases} -x & \text{if } x < 0 \\ x^2 & \text{if } 0 \leq x < 1 \\ 2x - 1 & \text{if } x \geq 1 \end{cases}$$

Determine algebraically whether or not $f(x)$ is differentiable at

1) $x = 0$ (6)

2) $x = 1$ (6)

QUESTION 8**10 MARKS****Solve the following Integrals.**

a) (5)

$$\int \left(\frac{1}{3}x^3 + 3x^2 + 5x + 1 \right) dx$$

b) (5)

$$\int_0^2 (x^2 + 1) dx$$